ITA03 · MOBILE COMPUTING

Assignment 2

Mobile Telecommunication Systems and Multiple Access

DATE:09/11/26 **NAME: Bharani. R.P**

REG NO:1925L260003

Executive summary

This report answers Q1–Q7 and includes working seeded simulations, measured tables and plots, a source-code appendix and a companion raw-results CSV. The model evidence supports four practical conclusions. First, wireless access does not by itself provide mobile computing: application state, local storage, synchronization and conflict handling matter when links fail. Second, equal nominal FDMA/TDMA channel counts do not mean equal effective capacity: guard/control assumptions and whether channels are reserved across idle time dominate. Third, synchronous equal-power Walsh CDMA stays orthogonal in the ideal model, while random codes and asynchronous near-far interference raise errors. Fourth, OFDMA best-subcarrier scheduling raises sum rate but gives some users zero resources in individual realizations; a minimum-rate constraint protects each user with almost no sum-rate loss in this experiment.

The Assignment 2 experiments use the fixed seed 7,312,026. The OFDMA comparison averages 1,000 Rayleigh-fading realizations. The generated numbers are results of stated simplified models, not standards compliance tests or field measurements. The separate Python file reproduces all recorded rows in the companion CSV.

1. Model assumptions and terminology

Mobile communication is the exchange of information with a moving or nomadic endpoint. Mobile computing includes the device, software, data and services that let a user compute while moving, disconnected, or operating under resource and context constraints. Wireless describes a link medium; mobile describes mobility. The terms overlap but are not synonyms. The experiment-specific parameters below are engineering assumptions, distinct from standardized GSM/DECT values cited later.

Model	Assumption
Random seed	7,312,026 for all experiments; Q1 runs availability-specific deterministic streams.
Q1	10,000 meter reports, one generated per minute; independent per-minute link availability; queued records sent in order at the first available minute.
Q2	5,000 one-minute epochs. FDMA has 76 usable circuits after assumed guard/control deductions; TDMA has 70 user slots after one of eight slots/carrier is reserved. For telemetry, the FDMA baseline reserves a channel for a 10-epoch transaction window; TDMA releases its slot after the one-epoch burst.
Q3	BPSK, 5,000 bits per case, chip lengths 8/16/64, matched-filter despreading, AWGN $\sigma=0.25/\sqrt{L}$ at decision output. "Near-far" uses one user 10× in power and a half-chip timing-offset correlation model.
Q4/Q7	8 users, 64 tones, SNR 15 dB, unit-mean exponential power gains (Rayleigh fading), Shannon rate proxy $\log_2(1+\text{SNR}\cdot\text{gain})$, 1,000 independent realizations. Redesign floor: 12 b/s/Hz/user per realization.
Q5	Uniform linear array with half-wavelength element spacing, ideal isotropic elements, broadside steering, equal powers and no noise/multipath. SIR target 10 dB.
Q6	Five criteria scored 1–5 with weights 25%, 20%, 20%, 15%, 20%; weights were selected for the four use cases and held fixed for baseline ranking.

Measurements compare the same traffic/channel realization where the question calls for a comparison. The code writes 40,430 raw rows. Channel counts and performance values are model-level quantities; protocol overhead beyond the stated assumptions is not inferred.

2. Q1 – Mobile communication or mobile computing

2.1 Classification criteria and predictions

I classify each case along five axes: (1) whether the user/device changes location during use; (2) whether the access link is wireless; (3) whether the application must survive disconnection; (4) whether location/context affects decisions; and (5) whether computation or data must be available locally under device limits.

Case	Classification	Main problem and reason
(a) Rugged tablet over cellular data	Mobile + wireless	Both. Cellular access and roaming are mobile communication; work-order caching, state, retries and synchronization are mobile computing.
(b) Laptop on Ethernet in changing offices	Nomadic, fixed-wired at each office	Primarily ordinary network communication and portable computing. The user moves between fixed connections; the link is not wireless.
(c) Fixed wireless router at remote substation	Fixed-wireless	Primarily mobile communication/link engineering (radio availability, antenna, backhaul). The router is not mobile merely because its link is wireless.
(d) Push-to-talk handset	Mobile + wireless	Primarily mobile communication: group call setup, coverage, latency, handover and prioritization.

		Device software remains relevant but is not the main challenge.
(e) Offline meter-reading tablet then sync	Mobile computing, intermittently connected	Local capture, durable queue, timestamps, idempotent upload and conflict resolution define correctness even when no link exists.
Additional ambiguous case: tablet docked in a moving service van, GPS logged locally, Wi-Fi at depots and cellular upload only after each job	Mobile/nomadic endpoint; sometimes wireless, often disconnected	It moves physically, but access is intermittent and app work is local. Treat as mobile computing with opportunistic mobile communication; docking does not remove offline requirements.

2.2 Simulation: immediate send versus store-and-forward

For each of 10,000 minute-indexed readings, the immediate strategy attempts delivery once and loses it if the link is down. Store-and-forward retains the reading and retries at the first available minute. The reported mean delay is conditional on delivered queued records; records still queued at the simulation horizon are counted as undelivered at the horizon.

Link availability	Immediate delivered	Immediate lost	Queued delivered	Delayed queued records	Queued at horizon	Mean queue delay (min)
90%	9,007	993	10,000	993	0	0.111
60%	6,058	3,942	9,999	3,941	1	0.660
30%	3,045	6,955	9,994	6,949	6	2.329

At 30% availability, immediate delivery loses 6,955 of 10,000 reports, while the queue delivers 9,994 and leaves six pending at the finite horizon. The link alone cannot provide persistence, ordering, deduplication or reconciliation. Offline meter capture and cellular work-order clients need application-level changes. Push-to-talk instead depends more on network-level call continuity and handover. The laptop benefits from portable computing but does not need radio adaptation in the stated setup.

3. Q2 – Sharing a channel: FDMA versus TDMA

3.1 Capacity and overhead estimate

Both options advertise 80 nominal user channels over 2 MHz per direction. FDMA: 2,000 kHz/25 kHz=80. For this hypothetical comparison, reserve two channel-width equivalents for edge guard and two for control/measurement, leaving 76 usable circuits (95%). TDMA: 2,000/200=10 carriers; 10×8=80 nominal slots. Reserve one slot on each carrier for synchronization/control, leaving 70 user slots (87.5%). An additional assumed 6% guard/synchronization payload loss leaves 70×0.94=65.8 full-slot-equivalent payload units; it reduces bits per slot, not the count of voice circuits. These overhead values are explicit assumptions, not GSM standard overhead values.

For real GSM context, ETSI GSM 05.02 specifies a TDMA radio structure; the classic frame is about 4.615 ms with eight time slots per carrier, while a GSM carrier is 200 kHz. The course scenario's 200 kHz × 8 slot arrangement resembles that structure, but the 2 MHz budget, 76/70 usable allocations and guard deductions here are assumed for the assignment model [1].

3.2 Traffic predictions and simulation design

Dataset	Designed load / edge case	Prediction
Continuous voice	90 long-lived users with on/off persistence over 5,000 epochs	Both saturate; FDMA's 76 circuits should block less than TDMA's 70 slots.
Bursty telemetry	500 sources, about 10% report activity; FDMA reserves for a 10-epoch transaction while TDMA releases a slot after each one-epoch packet	TDMA should use the resource pool much more efficiently because inactive reservation time is reclaimed.

70/30 voice–data mix	70 persistent voice sources and 30 burst data sources; same transaction reservation rule for data	TDMA should reduce blocking by reclaiming short packet resources.
Synchronized edge	85 simultaneous one-epoch reports every 20 epochs	Neither can serve all requests; fewer TDMA slots should increase burst blocking.

The simulation records requested and served channel-epochs, blocked channel-epochs, utilization (occupied/usable capacity) and wasted channel-epochs. The reservation rule deliberately separates circuit-switched holding from packet-slot assignment; without that assumption, equal request epochs would make both schemes differ mainly by their usable channel counts.

Traffic case	FDMA util.	FDMA waste	FDMA block	TDMA util.	TDMA waste	TDMA block
Continuous voice	100.00%	0.00%	10.86%	100.00%	0.00%	17.89%
Bursty telemetry	100.00%	0.00%	84.07%	68.13%	31.87%	0.00%
70/30 voice-data	99.32%	0.68%	20.76%	98.29%	1.71%	1.03%
Synchronized edge	100.00%	0.00%	10.59%	100.00%	0.00%	17.65%

Under the modeled reservation policy, burst-data blocking is 84.07% for FDMA versus effectively 0% for TDMA; in the 70/30 mix it is 20.76% versus 1.03%. Continuous voice reverses the comparison: 10.86% FDMA blocking versus 17.89% TDMA due to the 76 versus 70 capacity difference. The synchronized edge blocks 10.59% and 17.65%, respectively. This is why nominal channel count is not a measure of usable capacity: the result depends on guard/control loss and resource holding behavior.

3.3 Campus telemetry decision

A 200-byte report each minute from each of 5,000 sensors is a small packet workload. Dedicated per-device circuits are not justified: the aggregate bit rate is only 5,000×200×8/60≈133,333 bit/s (about 133 kbit/s before headers), but 5,000 long-lived channel reservations would consume the radio pool even when sensors are silent. Use scheduled/shared packet access with batching, sleep states and retry/backoff. A dedicated FDMA or TDMA circuit is reasonable only for a small set of latency-critical alarms or control endpoints.

4. GSM architecture, services and procedures

4.1 Architecture and radio interface

A GSM system separates the Mobile Station (ME plus SIM), Base Station Subsystem (BTS and BSC), and Network Switching Subsystem. The MSC switches calls and coordinates mobility; a VLR caches visiting subscriber/location data; the HLR stores subscription and routing information; the AuC supplies authentication material; the EIR can check equipment identity; and a GMSC provides interconnection to external networks. Operations and maintenance functions supervise the network. Main interfaces are Um (MS–BTS radio), Abis (BTS–BSC) and A (BSC–MSC). GSM 03.02 describes functional network entities and subscriber/location roles [2].

On the Um interface, GSM uses FDMA carrier selection with TDMA time slots. The classic GSM carrier spacing is 200 kHz; each TDMA frame has eight time slots and lasts about 4.615 ms. A physical channel follows a timeslot on a carrier; logical channels map traffic and signaling onto those physical resources. The precise burst/frame structure is specified by GSM 05.02 [1].

4.2 Services, localization and call flow

Services include teleservices such as telephony and short message service, bearer services for data transport, and supplementary services such as call forwarding, barring, waiting and caller identity. For a mobile-originated call, the MS first requests access on RACH; the network returns an assignment on AGCH; signaling may use SDCCH for identity/authentication/call setup; then the BSC assigns a traffic channel (TCH). For a mobile-terminated call, the GMSC queries the HLR for routing information, the serving MSC pages the current location area on PCH, the MS responds, and the network assigns signaling and traffic resources.

Localization is hierarchical: the MS camps on a cell, registers or updates its location area (LA), and the VLR/HLR maintain enough current routing context to page and route incoming calls. Cell reselection while idle differs from handover during an active connection. Handover decisions use radio measurements such as received level/quality and neighbor-

cell conditions; intra-BSC handovers can be coordinated by the BSC, while inter-BSC/inter-MSC cases require more network signaling and resource reservation.

4.3 Security and extensions

Classic GSM authenticates the subscriber using SIM-held secret material and a network challenge/response; the network can cipher the radio link. Temporary identities (TMSI) reduce routine exposure of the permanent IMSI. Security is not equivalent to end-to-end encryption: legacy GSM has no mutual authentication in its original form, and ciphering protections depend on the selected algorithm and network configuration. GSM security functions and algorithm roles are specified in the GSM/3GPP security series [3].

GPRS adds packet data service with packet control and serving/gateway support (SGSN/GGSN); radio resources can be scheduled for packet bursts rather than holding a circuit for an entire data session [4]. HSCSD is a circuit-switched data extension that can allocate multiple GSM timeslots to one connection, increasing rate at the cost of holding more circuit resources [5]. Classic DECT is a cordless access system combining multiple RF carriers with TDMA/TDD; the ETSI overview describes 24 full-slot positions in a 10 ms frame and typical multi-carrier operation [6]. These extensions illustrate that system architecture and resource scheduling evolve with service patterns.

5. Q3 – CDMA: orthogonality, spreading and near-far

5.1 DS-CDMA model and predictions

Map bit $b \in \{-1,+1\}$ to BPSK. For spreading code $c_u[k] \in \{-1,+1\}$ of length L, user u sends chips $x_u[k]=\sqrt{P_u}\, b_u\, c_u[k]$. The base station sums users and noise. A matched filter/despreader computes $z_u=(1/L)\Sigma_k\, y[k]c_u[k]$ and decides $\hat{b}_u=\text{sign}(z_u)$. Processing gain is Gp=L chips/bit, or $10\log_{10}L$ dB. Synchronous Walsh/Hadamard rows satisfy $(1/L)c_u^{T}c_v=0$ for $u \neq v$.

Before running the simulation, I predicted: (i) synchronous Walsh users with equal timing should have near-zero multiuser interference and BER limited mainly by noise; (ii) random non-orthogonal codes create multiple-access interference, so BER rises as user count grows, though increasing L reduces typical cross-correlation; (iii) a 10×-power user should still be separable under perfect synchrony and a flat channel, but a half-chip timing error mixes shifted chip correlations and should produce severe near-far errors. Thus “unequal power alone breaks orthogonality” is not correct under the ideal synchronous model; timing error and multipath destroy the orthogonal condition.

5.2 Measured results

L chips	Processing gain	K at full load	Walsh sync BER	Random-code BER	10× power + offset BER	Offset max |corr|
8	9.03 dB	8	0.0000	0.2506	0.4986	0.50
16	12.04 dB	16	0.0000	0.1242	0.5098	0.50
64	18.06 dB	64	0.0000	0.1476	0.5020	0.50

With L=64 random codes, the simulated maximum user count meeting BER≤1% was 15 in the tested finite-seed sweep. This is not a general CDMA capacity formula: support depends on target BER, code realization, power control, noise, timing, fading and receiver type. Power control is essential because received powers should be nearly equal; otherwise a strong nearby user can overwhelm a weak user after despreading. Multipath produces delayed replicas with nonzero cross-correlation even when codes are orthogonal at zero lag.

6. Q4 – OFDMA allocation

6.1 Spacing, symbol duration and orthogonality

For useful symbol duration T_u, choose subcarrier spacing $\Delta f = 1/T_u$. With 15 kHz spacing, $T_u = 66.67$ μs. A cyclic prefix copies the end of the useful symbol before transmission; if its duration exceeds the channel's significant delay spread, it absorbs multipath delay and makes the channel convolution appear circular over the FFT window. The prefix consumes time and reduces payload efficiency.

Numerical check: for 64 useful samples, the normalized inner product between subcarriers k and ℓ is $(1/64)\Sigma_n \exp[j2\pi(k-\ell)n/64]$. For $k, \ell \in \{0,1,2\}$, the maximum off-diagonal magnitude measured by the code is about 6.7×10^{-17}, numerical round-off around zero. Orthogonality requires synchronized FFT timing and sufficiently small residual frequency offset.

6.2 Allocation methods, predicted behavior and experiment

Fixed contiguous blocks give each of eight users eight adjacent tones. Interleaved round-robin gives each user every eighth tone. Best-subcarrier assigns each tone to the user with the largest instantaneous Shannon-rate proxy on that tone. The first two should be fair and similar under independent identically distributed fading; best-subcarrier should maximize

sum rate but can allocate zero tones to a weak user in a realization. Each method was run on the same 1,000 seeded 8×64 fading matrices.

Allocation	Mean sum rate	Mean rate/user	Lowest user long-run mean	Jain / realization	Worst realized user	Min tones
Fixed contiguous	277.23	34.65	34.47	0.986	15.96	8
Interleaved	277.17	34.65	34.47	0.986	17.11	8
Best-subcarrier	403.75	50.47	49.59	0.904	0.00	0
Min-rate redesign	402.95	50.37	49.74	0.917	12.18	2

Best-subcarrier mean sum rate was 403.75 b/s/Hz versus 277.23 for fixed contiguous, a 45.6% gain. But best-subcarrier assigned zero tones to at least one user in some realization, and its mean per-realization Jain index fell to 0.904. Fixed and interleaved schemes each allocate eight tones to every user; their mean Jain index was about 0.986. Because channel gains are statistically symmetric, long-run average rate across 1,000 realizations looks almost equal even when a single realization starves users. This difference explains why per-realization fairness is the meaningful metric here.

As user count rises, channel-aware scheduling has more choices per tone: the sum-rate advantage over fixed allocation grows, while per-realization fairness declines. The 1,000-realization results are:

Users	Fixed sum rate	Best-tone sum rate	Best-tone Jain index
2	276.68	333.23	0.984
4	277.49	373.95	0.956
8	276.94	403.43	0.903
16	277.12	426.48	0.815

This is multiuser diversity coupled with a larger scheduling imbalance risk.

6.3 Lecture-hall decision

Choose a constrained channel-aware scheduler with a per-user minimum rate of 12 b/s/Hz per realization, then assign remaining tones to the largest instantaneous rate. The 12 b/s/Hz floor is deliberately below the fixed-block scheme's worst observed 8-tone rate (about 15.96 b/s/Hz), so it is a feasible minimum service target in this 15 dB model. The scheduler uses channel knowledge and a fairness constraint; it does not promise 12 b/s/Hz if the physical channel cannot meet the floor under real noise/interference. In deployment, admission control or a lower coding rate must preserve the guarantee.

7. Q5 – SDMA: reusing a channel in different directions

7.1 Array-factor derivation

For an N-element uniform linear array with spacing d and steering direction θ_0, the phase difference between adjacent elements toward θ is $\psi=(2\pi d/\lambda)(\sin\theta-\sin\theta_0)$. Summing N equal-amplitude element responses and normalizing to the steered main-beam peak gives $AF(\theta;\theta_0)=(1/N)\cdot\sin(N\psi/2)/\sin(\psi/2)$, with power gain $|AF|^2$. For $d=\lambda/2$ and broadside $\theta_0=0$, $\psi=\pi\sin\theta$. The first null satisfies $N\psi/2=\pi$, so $\sin\theta_{null}=2/N$ when $N\geq 2$; beamwidth contracts approximately as 1/N. Uniform weighting has appreciable sidelobes; tapering can reduce them at the cost of a wider main lobe.

The model sets desired B-beam gain to one, no noise and equal user powers, so when A's beam points at A and user B is separated by Δθ, $SIR_B\approx 1/|AF(\Delta\theta)|^2$. A 10 dB SIR requirement means $|AF|^2\leq 0.1$. This is an optimistic two-user isolation metric, not a full multiuser uplink capacity result.

7.2 Computed array results

Elements N	Half-power beamwidth	First-null beamwidth	Peak sidelobe	Minimum separation for 10 dB SIR
2	60.02°	180.00°	no_visible_sidelobe	53°
4	26.34°	60.00°	-11.303337791201058	23°
8	12.82°	28.96°	-12.797347896865576	11°

16	6.36°	14.36°	-13.14683680190688	6°

The following table gives the modeled SIR for user B at representative separations; the CSV records every 5° from 5° to 60° for each array size.

N elements	5°	15°	30°	60°
2	0.1 dB	0.7 dB	3.0 dB	13.6 dB
4	0.4 dB	4.0 dB	300.0 dB	14.4 dB
8	1.8 dB	29.1 dB	300.0 dB	17.9 dB
16	8.6 dB	29.2 dB	300.0 dB	36.9 dB

At separations below the threshold angle, the beam steered at A still illuminates B too strongly for 10 dB isolation. As N grows, the angle threshold falls from 53° (N=2) to 6° (N=16). At an exact pattern null the ideal model predicts unbounded SIR, but multipath, calibration error, mutual coupling, finite precision, angular spread and non-isotropic elements fill the null.

7.3 Limits and toll-plaza decision

Two users arriving from nearly the same direction have almost identical steering vectors at a single array, so spatial filtering cannot distinguish them reliably. Angular spread, moving reflectors, limited aperture, channel estimation error and sidelobes also reduce separation. At a toll plaza, vehicles queue and approach in a narrow angular cluster; I would not rely on SDMA as the sole capacity mechanism. Use sectors and time/frequency scheduling, with SDMA only when measured angular separation and channel rank support it.

8. Q6 – Choosing access methods for four deployments

8.1 Criteria, weights and scoring

Scores are 1 (poor fit) to 5 (strong fit). Spectral efficiency receives 25%; fairness/QoS 20%; interference or jamming robustness 20%; energy/cost 15%; and scalability 20%. The weights reflect the operating priority of each use case. Ratings are reasoned engineering judgments informed by the Q2 traffic results, Q3 BER/correlation tests, Q4 throughput/fairness measurements and Q5 beam separation—not universal technology rankings. Weighted score is Σ(weight×rating), on a 1–5 scale.

Wide-area cellular voice

Technique	Spectral	Fair/QoS	Robust	Energy/cost	Scalable	Weighted
SDMA	4	3	3	3	4	3.45
TDMA	3	4	3	4	3	3.35
FDMA	2	4	4	3	2	2.95
CDMA	4	4	2	2	4	3.30
OFDMA	5	4	3	2	5	3.95

Jam-resistant tactical link

Technique	Spectral	Fair/QoS	Robust	Energy/cost	Scalable	Weighted
SDMA	3	3	2	2	3	2.65
TDMA	2	3	2	4	2	2.50
FDMA	1	3	1	3	1	1.70
CDMA	4	3	5	3	4	3.85
OFDMA	4	3	2	2	4	3.10

High-rate stadium downlink

Technique	Spectral	Fair/QoS	Robust	Energy/cost	Scalable	Weighted
SDMA	4	3	3	3	4	3.45
TDMA	3	3	3	3	3	3.00
FDMA	2	2	4	3	2	2.55
CDMA	3	3	2	2	3	2.65
OFDMA	5	4	4	3	5	4.30

5,000 low-rate sensors

Technique	Spectral	Fair/QoS	Robust	Energy/cost	Scalable	Weighted
SDMA	3	3	3	2	3	2.85
TDMA	3	4	3	5	4	3.70
FDMA	2	3	4	3	2	2.75
CDMA	4	3	3	2	4	3.30
OFDMA	4	4	3	2	4	3.50

8.2 Recommended choices and sensitivity

Deployment	Primary / hybrid	Evidence and trade-off
Wide-area cellular voice	OFDMA primary; SDMA where radio separation permits	Highest score 3.95. Flexible scheduling and reuse; requires mobility-aware QoS and careful voice latency. Legacy GSM instead combines FDMA+TDMA.
Jam-resistant tactical link	CDMA/spread spectrum plus frequency agility and power control	Highest score 3.85. Spreading helps against narrowband interference; Q3 shows synchronization and near-far remain critical. Frequency agility is a supporting anti-jam measure, not one of the five scored access methods.
High-rate stadium downlink	OFDMA + SDMA	Highest score 4.30. Q4 shows 45.6% sum-rate gain from channel-aware scheduling; Q5 supports beam reuse when users are angularly separated. Keep

		minimum-rate/fairness constraints.
5,000 low-rate sensors	Scheduled TDMA with FDMA carrierization and sleep/batching	Highest score 3.70. Small packets can be scheduled with low radio-on time; Q2 shows packet-slot reuse avoids long idle reservations. Include random-access control for alarms.

Sensitivity check: each criterion weight was changed independently by −20% and +20%, then all weights renormalized. The winning technique did not change for any of the 40 perturbations: OFDMA for voice and stadium, CDMA for tactical, TDMA for sensors. This local sensitivity check does not prove global robustness to different rating assumptions. Real systems combine methods because they solve different dimensions: GSM combines FDMA carrierization with TDMA slots; modern systems pair OFDMA scheduling with SDMA/MIMO; sensors can combine frequency channels with scheduled time slots.

The least robust choice is CDMA without tight synchronization and power control: Q3's 10×-power half-chip-offset case approached 50% BER at high load. SDMA is also fragile when users are angularly clustered, as the toll-plaza case shows.

9. Q7 – Diagnose, redesign and validate

9.1 A. Bottleneck diagnosis

I selected best-subcarrier OFDMA as the design to improve. The original scheduler allocates each tone to the user with the highest instantaneous rate and has no minimum service constraint. Its computational workload is 64×8 rate comparisons per realization, but the important defect is allocation concentration: in the 1,000-realization experiment, at least one user received zero tones in some realizations and the worst user rate was zero. The mean sum rate is high, but a student with a poor channel can miss the entire scheduling interval.

9.2 B. Measurable objective

Preserve at least 12 b/s/Hz per user in each feasible realization, achieve zero floor violations in the same 1,000 channels, and keep mean sum-rate loss below 1% versus the unconstrained best-tone scheduler. The floor is the lecture-hall minimum service target selected in Q4.

9.3 C. Revised strategy

The redesign first reserves the strongest remaining tones needed to bring each user to the 12 b/s/Hz floor. It then assigns all remaining tones greedily to the user with the largest rate on each tone. The strategy reuses instantaneous channel-state information already needed by best-tone scheduling. It adds per-user accumulated-rate state and a feasibility check; it can reduce peak sum rate if many users have poor channels.

MIN_RATE_OFDMA(gain[users][tones], target=12):

compute rate[u,s] = log2(1 + SNR * gain[u,s])

remaining = all tones; allocation[u] = empty; rate_sum[u] = 0

while any rate_sum[u] < target:

choose an unmet user (prioritize weakest feasible user)

assign that user's best remaining tone; update its rate_sum

if no tones remain before all users meet target: declare infeasible

assign each remaining tone to the user with the highest rate on it

return allocation, per-user rates, sum rate, fairness

9.4 D. Prediction before the validation run

Prediction: the revised scheduler will eliminate zero-rate users and raise per-realization fairness; the mean sum rate will fall slightly because some tones go to users who are not the instantaneous best on those tones. A successful redesign should meet the 12 b/s/Hz floor in every tested realization while keeping the sum-rate penalty under 1%.

9.5 E. Same-condition validation results

Measure	Original best-subcarrier	Minimum-rate redesign	Change
Mean sum throughput (b/s/Hz)	403.75	402.95	-0.20%
Mean Jain fairness per realization	0.9035	0.9173	+0.0138

Worst user rate in any realization (b/s/Hz)	0.00	12.18	Floor protected
Minimum tones assigned to any user	0	2	No user left unallocated
12 b/s/Hz floor violations	Not constrained	0 across 1,000 realizations	Objective met

Mean sum rate fell from 403.75 to 402.95 b/s/Hz, a 0.20% loss—within the 1% target. Average per-realization Jain fairness increased by 0.0138; the worst user received 12.18 b/s/Hz, and there were zero floor violations. The prediction was supported. The original remains preferable when users have similar service priority and sum throughput is the only objective, or when a minimum rate is infeasible because channels are too poor or too few. The redesign depends on timely CSI and adds per-user rate bookkeeping; the assignment algorithm should rotate priority or use weakest-first selection to avoid order bias.

9.6 F. Final defense

The redesign addresses the diagnosed resource concentration directly rather than replacing OFDMA. It met the measurable service floor and retained 99.80% of the original sum throughput in this simulation. Its strongest practical use is a classroom, where each student needs a minimum service rate. For a bulk-download event with no per-user guarantee, the original best-tone scheduler can remain useful, subject to admission control and starvation monitoring.

Appendix A Reproduction and submitted files

The separate file ITA03_Assignment_2_Simulations.py contains all Q1–Q7 simulation functions and writes the raw long-form measurements to ITA03_Assignment_2_Raw_Results.csv. Run it with Python 3 from the folder containing an outputs subfolder; it uses only the Python standard library. The fixed seed is 7,312,026. The CSV includes per-case outcomes, per-user throughput observations for the 1,000 OFDMA channel realizations, CDMA BER/correlation/processing gain, array-pattern samples and Q6 scores/sensitivity outcomes.

Reproducibility command from the project folder: python outputs/ITA03_Assignment_2_Simulations.py. The reported Monte Carlo tables are generated by this source. Analytical equations and standards-derived parameters are identified separately from simulation assumptions.

Appendix B Academic integrity and AI disclosure

AI assistance was used to organize the report, draft explanations and implement the experiment source. ETSI/3GPP standards and cited textbooks were consulted for technical context. The submitting student should review the source, rerun the simulations, understand every assumption and be prepared to explain the design and results in a viva, following the course and institution's disclosure rules.